

A close-up, high-resolution photograph of an elderly person's face, focusing on their eyes. The skin is wrinkled, and the eyes are light-colored and looking slightly to the right. The lighting is soft, highlighting the texture of the skin.

CALMA

CLINICAL TRIAL: ALZ-AGITATION

Hope in every step toward Alzheimer's Care

A Phase 2, Multi-Center, Double-Blind, Randomized, Placebo-Controlled,
Trial of the Safety and Efficacy of IGC-AD1 on Agitation in Participants with
Dementia due to Alzheimer's Disease.

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Study Design

CALMA is a multicenter, randomized, double-blind, placebo-controlled study. The key criteria for inclusion is clinically significant agitation. This is assessed by a score of ≥ 4 in the Agitation subscale on the Neuropsychiatric Inventory-12 (NPI-12) during screening. Patients must also meet the International Psychogeriatric Association (IPA) agitation criteria and have excluded other potential medical causes of agitation.

Target Population

Participants with mild to severe dementia due to Alzheimer's Disease (AD), who meet the criteria for probable AD according to the NIA-AA criteria and for **clinically significant** agitation.

Enrollment (Estimated)

164 participants.

Dosing



Treatment

42 days plus 2 days of taper down at the end. Estimated total time in the study (including pre-screening, screening, treatment, and follow-up): 50-57 days.

Intervention

IGC-AD1 is an oral solution containing 2 active ingredients: delta-9 tetrahydrocannabinol (THC) and melatonin.

[ClinicalTrials.gov ID: NCT05543681](https://clinicaltrials.gov/ct2/show/study/NCT05543681)



Day 1 and Day 2: QD (1 mL/day)



Day 3 to Day 42: BID (1 mL twice a day)



Day 43 and Day 44: QD (1 mL/day)

Primary Objective & Endpoint

Objective: Assess the efficacy of IGC-AD1 compared to placebo in Participants with Agitation in dementia due to AD, as assessed by the Cohen-Mansfield Agitation Inventory (CMAI) after 6 weeks of treatment.

Endpoint: Difference in CMAI mean scores from baseline to week 6.

Secondary Objective & Endpoint

Objective: To assess acute efficacy of IGC-AD1 on symptomatological Agitation, in dementia due to AD, in a two-week placebo-controlled trial using CMAI.

Endpoint: Change in mean CMAI between baseline and week two.

Exploratory Objectives

Exploratory objectives include examining:

- Changes in the Mini-Mental State Examination, Cornell Scale for Depression in Dementia (CSDD), Quality of Life in Alzheimer's Disease (QOL-AD), Zarit Burden Interview (ZBI), Neuropsychiatric Inventory (NPI-12), and Clinical Global Impression (CGI).

- Changes in the use of psychotropic drugs or the need for rescue medication.
- Number of participants with polymorphisms in the apolipoprotein E (ApoE) gene, specifically ApoE ε4, ApoE ε3, and ApoE ε2, and their impact on each of the Neuropsychiatric Symptoms (NPS) measured at weeks 2, 4, and 6.
- Sparse Pharmacokinetic (PK).
- The impact of cytochrome P450 (CYP450) polymorphisms, specifically CYP2C9, on the efficacy of IGC-AD1 on each of the NPS measured at weeks 2, 4, and 6, and in the PK.

Laboratory Evaluations

Clinical laboratory parameters will be evaluated at screening and EOT

- Serum chemistry: creatinine, blood urea nitrogen (BUN), glucose, sodium, potassium, chloride, bicarbonate, calcium, phosphorus, vitamin D, alkaline phosphatase (ALP), AST, ALT, total bilirubin, direct bilirubin, LDH, albumin, total protein, and lipid panel (total cholesterol, LDL, HDL, and triglycerides).
- Hematology: complete blood count (RBC, hemoglobin, and hematocrit, total WBC count with differential counts, platelets).
- Urinalysis and urine drug screen.

Other Testing

Genotyping, Pharmacokinetics and Blood Biomarkers.

Eligibility Criteria



Inclusion criteria:

1. Must have a Caregiver willing to comply with all required study procedures.
2. The caregiver must be known to the participant and capable of using electronic devices such as a cell phone or computer and able to learn how to take blood pressure readings.
3. Diagnosis of probable AD according to NIA-AA criteria.
4. Participants who cannot consent may have a Caregiver's consent provided. Accepting consent must be consistent with established practice at the site and jurisdiction.
5. Participants must consent to genotyping and pharmacokinetics.
6. Clinically significant Agitation assessed by:
 - a. NPI (Agitation) ≥ 4
 - b. IPA agitation criteria
 - c. Agitation not caused by another psychiatric disorder, suboptimal care, underlying medical condition, or the physiological effect of a substance.
7. Negative drug screen, except for benzodiazepines.
8. All medications used for behavioral symptoms should be consistent for at least six weeks before screening, with allowance for dose changes of up to 25%.
9. Women should have a low probability of becoming pregnant (postmenopausal or surgically sterile).

Exclusion criteria:

1. Prior adverse reaction to cannabinoids or to any component of Study Drug.
2. Unstable medical illness, including cardiovascular, hepatic, renal, respiratory, endocrine, neurologic, or hematologic disease, or any condition that may interfere with safety assessment.

3. History of seizures, schizophrenia, or bipolar disorder.
4. Has participated in an investigational drug or device study within 30 days prior to study start.
5. Positive urine drug screen, except for benzodiazepines.
6. History of alcohol and drug use disorder within one year prior to enrollment.
7. Hypertension: Participants with a history of uncontrolled hypertension with a hypertensive crisis in the six months before enrollment.
8. Falls: Participants with a history of more than two falls in the six-month period prior to enrollment and a history of falls resulting in injuries or associated with a new acute illness, loss of consciousness, fever, or abnormal blood pressure.

Follow up calls

An individual from the study team will conduct safety calls daily for the days 2, 3 and 4 and then every third day to review the entries in the logbook, changes of concomitant medication, solicited and non-solicited AEs, heart rate, blood pressure and weight and drug compliance.

Scales



Cohen-Mansfield Agitation Inventory (CMAI): The CMAI is a 29-item scale used to assess agitation, rated on a 7-point scale of frequency. The primary caregiver rates the frequency of the participant's physical aggression, physically nonaggressive, and verbally agitated behaviors. The total score ranges from 29-203, with scores ≥ 45 indicating clinically significant agitation.

Neuropsychiatric Inventory (NPI-12): The NPI-12 is a clinician-administered scale informed by the caregiver. It assesses hallucinations, delusions, agitation/aggression, dysphoria/depression, anxiety, irritability, disinhibition, euphoria, apathy, aberrant motor behavior, sleep and night-time behavior changes, and appetite and eating changes. It evaluates the frequency, severity, and caregiver distress. For screening, only the agitation subscale will be used. A score ≥ 4 in any subscale is considered clinically significant.

Zarit Burden Interview (ZBI): The ZBI is a Caregiver self-reported measure. Each item in the interview is a statement that the Caregiver is asked to endorse using a 5-point scale. Response options range from 0 (never) to 4 (nearly always). The total ZBI score ranges from 0 to 88, with higher scores denoting more stress experienced by Caregivers.

Clinical Global Impression (CGI): The Clinical Global Impression (CGI) scale, created in 1976 by the National Institute of Mental Health (NIMH), is a widely recognized tool for assessing a patient's overall functioning from the clinician's perspective, both before and after treatment initiation. This scale is instrumental in clinical settings for gauging the impact of therapeutic interventions. In our study we focus on CGI-Severity and CGI -improvement to assess agitation.

Schedule of events

Procedures	Pre-Screening	Screening and Enrollment	Safety call	Safety & Scales	Safety call	Safety & Scales Video call or in person	Safety call	EOT	Safety Follow up call EOT+4 days
Trial day = D	D minus 45	D0 Baseline	D2, D3, D4 D7, D10	D 14 (Wk. 2)	D17, D20, D23, D26	D28 (Wk. 4)	D31, D34, D37, D40	D42 (Wk. 6) EOT	D46 EOT+4
Presentation of the trial, NPI (Agitation ≥ 4) Send ICF, and agitation behaviors list for review.	X								
Sign Informed Consent Form		X							
CMAI		X		X		X		X	
Demographics		X							
Review Inclusion/Exclusion Criteria		X							
Review of Medical History/Interim medical history		X		X		X		X	
Dispense IGC-AD1		X							
Review of Prior or Concomitant Medications		X		X				X	
Assessment of changes in psychotropic drugs				X		X		X	
Review of Logbook, assess Adverse Events, compliance, vital signs, drug accountability			X	X	X	X	X	X	X
Collect Logbooks				X		X		X	X
Physical and Neurologic Examination		X		X				X	
Blood Draw		X						X	
PK blood draws*				X				X*	
Urinalysis		X						X	
12 lead ECG		X						X	
CGI-S(Agitation)		X		X				X	
CGI-I(Agitation)				X				X	
MMSE-2		X						X	
CSDD (Depression)		X		X		X		X	
NPI-12**		X		X		X		X	
QOL-AD		X		X		X		X	
ZBI Caregiver**		X		X				X	
SAS-SF***		X							
MOAA/S		X						X	

*Optional Blood samples for pharmacokinetics will be collected on Day 14 or 42. However, as the design follows a population PK approach, and the protocol specifies that analysis will be conducted for a subset of participants, the sponsor may communicate to the site not to proceed with the draws.

** NPI-12-D and ZBI-Caregiver are applied to non-professional Caregivers only.

*** The SAS-SF is self-administered by the Caregiver during the Baseline visit and repeated on day 7 in the logbook



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